TOC ANALYSER CALIBRATION WORKSHEETS

(Ref. MICLAB 115)

Inorganic Carbon Standard Worksheet

Date, Name of preparer		line 1
Lot of Na_2CO_3 used in prep.		line 2
Weight of Na_2CO_3 + weigh boat		line 3
Weight of weigh boat		line 4
Weight of Na_2CO_3		line 5

Table 1. **Preparation of the Stock Na_2CO_3 standard**

Concentration of stock solution	nominal 500 ppm	line 6

Table 2. **Preparation of the Dilute Na_2CO_3 standard**

Concentration of dilute solution	nominal 25.0 ppm	line 7

TOC ANALYSER CALIBRATION WORKSHEETS

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Organic Carbon Standard Worksheet

Date, Name of preparer		line 1
Lot of KHP used in prep.		line 2
Weight of KHP + weigh boat		line 3
Weight of weigh boat		line 4
Weight of KHP		line 5

Table 1. **Preparation of the Stock KHP standard**

Concentration of stock solution	nominal 2000 ppm	line 6

Table 2. **Preparation of the Dilute KHP standard**

Concentration of dilute solution	nominal 25.0 ppm	line 7

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Calibration Worksheet for Autosampler Calibration

Date, Name of Analyst ____

Measurements of Standards

Average TC of 3rd DI blank		$=TC_{DI\ blank}$
Average TC of 3rd TC(KHP) standard		$=TC_{ave,TC}$
Average TC of 3rd IC (Na_2CO_3) standard		$=TC_{ave,IC}$
Average IC of 3rd IC (Na_2CO_3) standard		$= IC_{ave,IC}$
TC value of TC(KHP) standard from formulation		$= TC_{std}$
Adjusted TC value of the TC(KHP) standard = $TC_{std} + TC_{DI\ blank}$		$= TC_{adj,TC}$

Calculation of Calibration Constants

Current TC calibration constant		$=TC_{cal,old}$

New TC calibration constant $= TC_{cal,old} \bullet \frac{TC_{adj,TC}}{TC_{ave,TC}}$	$=TC_{cal,new}$

Current IC calibration constant		$=IC_{cal,old}$

New IC calibration constant $= IC_{cal,old} \bullet \frac{TC_{adj,TC}}{TC_{ave,TC}} \bullet \frac{TC_{ave,IC}}{IC_{ave,IC}}$	$=IC_{cal,new}$

Verification of Calibration

Average TC of 3rd TC (KHP) standard		$=TC_{ave,chk,TC}$
Average TC of 3rd IC(Na_2CO_3) standard		$=TC_{ave,chk,IC}$
Average IC of 3rd IC (Na_2CO_3) standard		$=IC_{ave,chk,IC}$
New $TC_{adj,TC} = TC(KHP)_{std}$ + average TC 3rd DI Blank =		
TC Std % difference $= \frac{(\text{New } TC_{adj,TC} - TC_{ave,chk,TC}) \times 100}{\text{New } TC_{adj,TC}}$ =		(NMT 3%)
IC Std % difference $= \frac{(TC_{ave,chk,IC} - IC_{ave,chk,IC}) \times 100}{TC_{ave,chk,IC}}$ =		(NMT 3%)

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500 ppb Sucrose Standard Worksheet

Date, Name of Preparer		line 1

Lot No. of Sucrose used		line 2

Carbon Assay (% value / 100)	nominal 0.421	line 3

Gross Weight (Sucrose + Boat)		line 4

Tare Weight of Weigh Boat		line 5

Nett Weight of Sucrose		line 6

Carbon Conc. of Stock Solution (ppb C $= \dfrac{\text{g [line 6]} * \text{ca [line 3]} * 10^6}{1\text{ L}}$)	nominal 25,000 ppb C	line 7

Carbon Conc. of Working Solution ([line 7] ÷ 50)	nominal 500 ppb C	line 8

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500 ppb Benzoquinone Standard Worksheet

Date, Name of Preparer		line 1

Lot No. of Benzoquinone used		line 2

Carbon Assay (% value / 100)	nominal 0.666	line 3

Gross Weight (Benzoquinone + Boat)		line 4

Tare Weight of Weigh Boat		line 5

Nett Weight of Benzoquinone	nominal 0.075g	line 6

Carbon Conc. of Stock Solution (ppb C = $\frac{\text{g [line 6]} * \text{ca [line 3]} * 10^6}{1 \text{ L}}$)	nominal 50,000 ppb C	line 7

Carbon Conc. of Working Solution ([line 7] $\div$ 100)	nominal 500 ppb C	line 8

TOC ANALYSER CALIBRATION WORKSHEETS

(Ref. MICLAB 115)

Suitability Verification Worksheet

Date and Name of Analyst		line 1
Calculated Concentration of Sucrose Standard		line 2
Calculated Concentration of Suitability Standard		line 3
Average TOC of Water Blank		line 4
Average TOC of Sucrose Standard		line 5
Average TOC of Suitability Standard		line 6
Response Efficiency $\text{Response Efficiency (\%)} = \frac{(TOC_{suitability} - TOC_{blank})}{(TOC_{sucrose} - TOC_{blank})} * \frac{C_{sucrose}}{C_{suitability}} * 100$	between 85% and 115%	line 7